THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Mushrooms obtain food from: 1

(a) sunlight **(b)** dead and decaying matter
(c) soil water **(d)** air

2. Variegated leaves have: 1

(a) no veins **(b)** no cells
(c) green and non-green patches **(d)** only starch

3. During photosynthesis, light energy is changed into: 1

(a) chemical energy **(b)** heat only
(c) sound **(d)** electrical energy

4. The upward movement of water is aided by: 1

(a) transpiration **(b)** respiration **(c)** germination **(d)** photosynthesis

5. Which gas do plants take in during respiration? 1

(a) hydrogen **(b)** nitrogen **(c)** helium **(d)** oxygen

6. Mirrors in a periscope are placed at: 1

(a) 0° to the tube **(b)** 90° to the tube **(c)** 30° to the tube **(d)** 45° to the tube

7. Light travels fastest in: 1

(a) water **(b)** air **(c)** glass **(d)** vacuum

8. A beam of light is made of: 1

(a) sound **(b)** shadows **(c)** one wave only **(d)** many rays

9. If you walk towards a plane mirror at 1 m/s, your image moves towards you at: 1

(a) 1 m/s (so closing speed is 2 m/s) **(b)** 0.5 m/s
(c) 0 m/s **(d)** 4 m/s

10. A rainbow-coloured pattern from sunlight is due to: 1

(a) reflection by plane mirror **(b)** shadow
(c) pinhole **(d)** splitting of white light

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What are saprotrophs? Give an example. 2

12. What is the role of Rhizobium in legumes? 2

13. What is a periscope used for? 2

14. How does a kaleidoscope form patterns? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three differences between day and night activities in a plant. 3

16. A student sees the word 'STOP' in a mirror. How does it look? Why? 3

17. Why must the leaf be boiled in alcohol before testing? 3

OR

How do leaves exchange gases? 3

18. How can you show that reflection obeys equal angles? 3

OR

Give uses of mirrors in daily life. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Explain the structure of a leaf. 5
(ii) Roles of veins.
(iii) Why flat?

OR

(i) How do plants make glucose? 5
(ii) What happens to excess?
(iii) Test it.

20. (i) How does the shadow of a person change from morning to noon? 5
(ii) Why?
(iii) Use in sundials.

OR

(i) A boy stands 2 m from a plane mirror. How far is his image from him? 5
(ii) He moves 1 m closer. New distance?
(iii) State why.

END OF QUESTION PAPER — Total: 40 Marks

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